Listing 11.5 `basic_nexus_planner.py` (`execute_plan`)

```
def execute_plan(self, nexus, agent, plan: Plan) -> str:
        context = {}
        plan = plan.generated_plan
        for task in plan["subtasks"]:     #1
            if task["function"] == "for-each":     #2
                list_name = task["args"]["list"]
                index_name = task["args"]["index"]
                inner_task = task["args"]["function"]

                list_value = context.get(list_name, [])
                for item in list_value:
                    context[index_name] = item
                    result = nexus.execute_task(agent, inner_task, context)
                    context[f"for-each_{list_name}_{item}"] = result

                for_each_output = [     #2
                    context[f"for-each_{list_name}_{item}"] ↪
                      for item in list_value
                ]
                context[f"for-each_{list_name}"] = for_each_output

                for item in list_value:     #3
                    del context[f"for-each_{list_name}_{item}"]

            else:
                result = nexus.execute_task(agent,
                                            task,
                                            context)     #4
                context[f"output_{task['function']}"] = result

        return context     #5
```

#1 Iterates through each subtask in the plan
#2 Handles functions that should be iterated over and adds full list of results to the context
#3 Removes individual for-each context entries
#4 General task execution
#5 Returns the full context, which includes the results of each function call

The returned context from the entire execution is sent in a final call to the LLM, which summarizes the results and returns a response. If everything goes as planned, the LLM will respond with a summary of the results. If there is an error or something is missing, the LLM may try to fix the problem or inform the user of the error.

Let’s now open Nexus again and test a planner in operation. Load up the same agent you used last time, but select the planner under the Advanced options this time, as shown in figure 11.8. Then, enter the goal prompt as you did before, and let the agent take it away.